ISE 573 — Manufacturing Execution Systems

Assignment: OEE Measurement, Reliability, and Queueing

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DUE on April 4

Instructions. Show all work. State every formula *before* substituting numbers. Box final numerical answers. No SQL or programming required — use only the definitions and formulas from the OEE lecture. **Total: 100 points.** Submit as a single PDF.

Name: _____ **Student ID:** _____ **Section:** _____

Key formulas (reproduced for reference)

OEE components

$$A = \frac{T_{\text{run}}}{T_{\text{planned}}}$$

$$P = \frac{N_{\text{actual}}}{T_{\text{run}}/t_{\text{ideal}}}$$

$$Q = \frac{N_{\text{good}}}{N_{\text{total}}}$$

$$\text{OEE} = A \times P \times Q$$

Reliability metrics

$$\text{MTBF} = \frac{T_{\text{run}}}{N_{\text{fail}}}$$

$$\text{MTTR} = \frac{\sum T_{\text{repair}}}{N_{\text{fail}}}$$

$$A_{\text{rel}} = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}}$$

$$R(t) = e^{-t/\text{MTBF}}$$

Queueing

Kingman (G/G/1):

$$W_q \approx \frac{\rho}{1-\rho} \cdot \frac{C_a^2 + C_s^2}{2} \cdot \frac{1}{\mu}$$

Little's Law: $L = \lambda W$

$$C^2 = \sigma^2/\bar{x}^2$$

Problem 1. OEE Computation from Shift Data — 20 pts

Reference: Lecture Parts 3 & 5 — OEE formula; Worked Example A

A horizontal machining centre ran a 10-hour day shift. The following data were recorded by the MES at end of shift.

Shift item	Value
Shift duration	600 min
Scheduled breaks + tool-crib time	40 min
Unplanned spindle motor failure	55 min
Changeover / first-piece inspection	25 min
Ideal cycle time	1.5 min/part
Actual parts produced	290 parts
Conforming (good) parts	279 parts

(a) State the planned production time T_p and run time T_r. Identify the Six-Loss category (L1–L6) for each downtime event.

(b) Calculate Availability A. Show formula and substitution.

(c) Calculate Performance P. State clearly what the denominator represents physically.

(d) Calculate Quality Q.

(e) Calculate OEE. State whether the result is *Unacceptable / Typical / Good / Excellent / World-class* and give the corresponding OEE percentage range from the benchmark table.

Problem 2. Six Big Losses — Classification and Pareto Priority — 20 pts

Reference: Lecture Part 4 — Six Big Losses; Part 7 — Pareto analysis

During a five-shift production week the MES logged the events below on a CNC turning cell. All timed entries are in minutes.

Event	Duration (min)	Reason code
Servo drive fault — machine halted	95	BD
Scheduled coolant flush / PM	45	PM
Product changeover (new part family)	70	CO
Insert indexing — repeated micro-stops	38	BD
Reduced feed rate: worn carbide insert	—	(speed loss)
First-off dimensional rejects (8 parts)	—	(quality)
Hydraulic leak repair	60	BD
Operator break overrun (unrecorded)	12	OT

Additional data: planned production time = 2 100 min; ideal cycle time = 2.0 min/part; actual output = 840 parts; speed-loss equivalent = 45 parts; good parts = 832.

(a) Assign each *timed* event to one of the Six Big Losses (L1–L6). State the OEE component (A, P, or Q) it reduces.

(b) Compute A, P, Q, and OEE for the week.

(c) Express each availability loss as a percentage of planned time. Construct a Pareto table: loss category, minutes, % of total losses, cumulative %. Which two categories account for more than 70% of total lost time?

(d) Based on the Pareto, state which management lever (TPM / SMED / Poka-yoke / quality improvement) should be applied first and justify your answer in one sentence.

Problem 3. MTBF, MTTR, Reliability Availability, and PM Interval — 20 pts

Reference: Lecture Part 2 — MTBF, MTTR, bathtub curve, PM interval formula

A laser-cutting machine was monitored over **15 production shifts (7 200 min total available time)**. The table shows each unplanned breakdown event.

Breakdown event	Repair duration (min)
Laser power supply fault	65
Assist-gas pressure loss	40
Chiller unit overtemperature	90
Motion-axis encoder fault	55
Laser power supply fault (repeat)	70

(a) Calculate MTBF in hours. Note: T_{run} = total available time − total repair time.

(b) Calculate MTTR in minutes and in hours.

(c) Calculate the reliability view of availability $A_{\text{rel}} = \text{MTBF}/(\text{MTBF} + \text{MTTR})$. Compare to the world-class target of 90% and state what the gap implies about the maintenance strategy.

(d) Using $R(t) = e^{-t/\text{MTBF}}$ and the criterion that PM should be performed before the probability of failure reaches 10%, calculate the recommended PM interval T_{PM} (in hours). How many 8-hour shifts does this correspond to?

(e) The repeat failure of the laser power supply in events 1 and 5 suggests the machine may be entering which region of the bathtub curve? Name the region and state the appropriate maintenance response.

Problem 4. Kingman's G/G/1 Formula and the Effect of Availability Loss on W_q 20 pts

Reference: Lecture Parts 0 & 9 — Kingman (1961), C^2, P-K vs. Kingman, Effects 1 & 2

A press-brake station receives jobs from an upstream laser cutter at average rate $\lambda = 10$ jobs/hr. The press-brake's nominal service rate is $\mu = 13$ jobs/hr. Interarrival variability is $C_a^2 = 1.4$ (upstream batch releases). Service variability is $C_s^2 = 0.7$ (controlled press cycle).

(a) Calculate utilisation ρ. Is the system stable? Explain in one sentence what happens to W_q as $\rho \to 1$.

(b) Apply Kingman's G/G/1 formula to compute W_q in minutes. Identify and state the three factors (utilisation factor, variability factor, time-scale factor) explicitly before multiplying.

(c) A 15% availability loss reduces the effective service rate to $\mu_{\text{eff}} = \mu \times A$. Compute ρ_{eff} and the new W_q under Kingman. By what factor does W_q increase?

(d) Breakdowns also inject variance into service time. Using

$$C_s^2(\text{eff}) \approx C_s^2(\text{nom}) + \frac{2\,\text{MTTR}}{\text{MTBF} \cdot t_s}$$

with MTTR = 50 min, MTBF = 600 min, and nominal cycle time $t_s = 60/13$ min, compute $C_s^2(\text{eff})$ and state how this result further inflates W_q.

(e) State in *one sentence* why Kingman's formula is preferred over the Pollaczek–Khinchine (P-K) formula for this shop-floor application.

Problem 5. C^2, Little's Law, and Integrative OEE–Queueing Analysis — 20 pts

Reference: Lecture Parts 0 & 9 — C^2, Little's Law, OEE diagnostic / Kingman predictor

A welding cell records the following interarrival times (minutes) for 8 consecutive jobs arriving from an upstream assembly station:

Job	1	2	3	4	5	6	7	8
Interarrival (min)	4	7	3	12	5	4	9	4

Service times at the welding cell are approximately constant at **5.5 min/job** (automated weld programme, minimal operator variation). The MES reports that over the past week the welding cell had **OEE = 0.72**, broken down as $A = 0.83$, $P = 0.91$, $Q = 0.95$.

(a) Compute the sample mean $\bar{x}$ and sample variance s^2 of the interarrival times. Then compute $C_a^2 = s^2/\bar{x}^2$. Using the C^2 landmark table from the lecture ($C^2 = 0$: deterministic; $C^2 < 1$: low variability; $C^2 = 1$: exponential; $C^2 > 1$: high variability), classify the arrival process.

(b) The service time is described as "approximately constant." State the value of C_s^2 and justify it from the C^2 definition. Compute nominal utilisation $\rho = \lambda/\mu$ using the mean interarrival time from part (a).

(c) Apply Kingman's formula using C_a^2 from part (a) and C_s^2 from part (b) to compute W_q (minutes). Then use Little's Law to compute the mean number of jobs waiting in queue, L_q.

(d) The OEE report shows $A = 0.83$. Using $\mu_{\text{eff}} = \mu \times A$, recompute ρ_{eff} and W_q under Kingman. By how many minutes does the queue wait increase relative to the no-availability-loss baseline from part (c)?

(e) Using the OEE–Kingman integration diagram from the lecture (*OEE as diagnostic / Kingman as predictor*), state which of the three management responses below is most appropriate for this cell and justify your choice using the numerical results from parts (b)–(d):

i. Improve maintenance to raise A (raise MTBF or lower MTTR)

ii. Add a parallel welding cell (add server, increase c)

iii. Throttle upstream release rate (reduce λ)

Problem	Topic	Points	Score
1	OEE computation from shift data	20	
2	Six Big Losses & Pareto	20	
3	MTBF, MTTR, PM interval	20	
4	Kingman's formula & availability loss	20	
5	C^2, Little's Law, integrative	20	
Total		**100**	